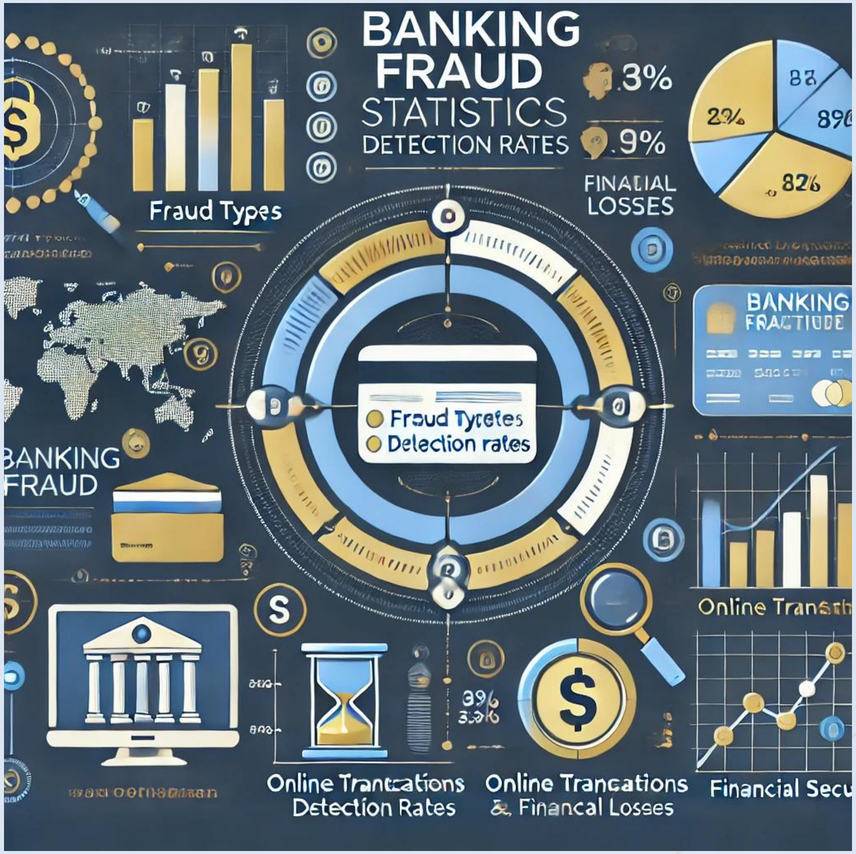
Banking Transaction Analysis

Detecting Fraud and Improving Transaction Efficiency

Presented by: Edwin Korir

Introduction

- **This project analyzes banking transactions to identify fraud, improve security, and optimize transaction efficiency using data-driven insights.**



Business Context

- Banking fraud and transaction failures cost financial institutions billions. This analysis aims to detect fraud patterns, assess transaction failures, and provide actionable recommendations for improving banking security.

Data Overview

- Dataset includes:
- - Transaction details (ID, Sender, Receiver, Amount, Type, Status)
- - Fraud indicators (Fraud Flag)
- - Geolocation & Device Data (Latitude, Longitude, Device Used)
- - Network characteristics (Latency, Bandwidth)
- - Security Data (PIN Code presence)

Process Steps

- 1. Data Cleaning & Preprocessing
- 2. Exploratory Data Analysis (EDA)
- 3. Fraud Detection Analysis
- 4. Transaction Success Rate Analysis
- 5. Visualizing Key Trends

Results & Business Application

- - Identified common fraud patterns and risk factors
- - Determined key causes of transaction failures
- - Found correlation between device type, network quality, and fraud
- - Provided data-driven recommendations for fraud prevention and system optimization

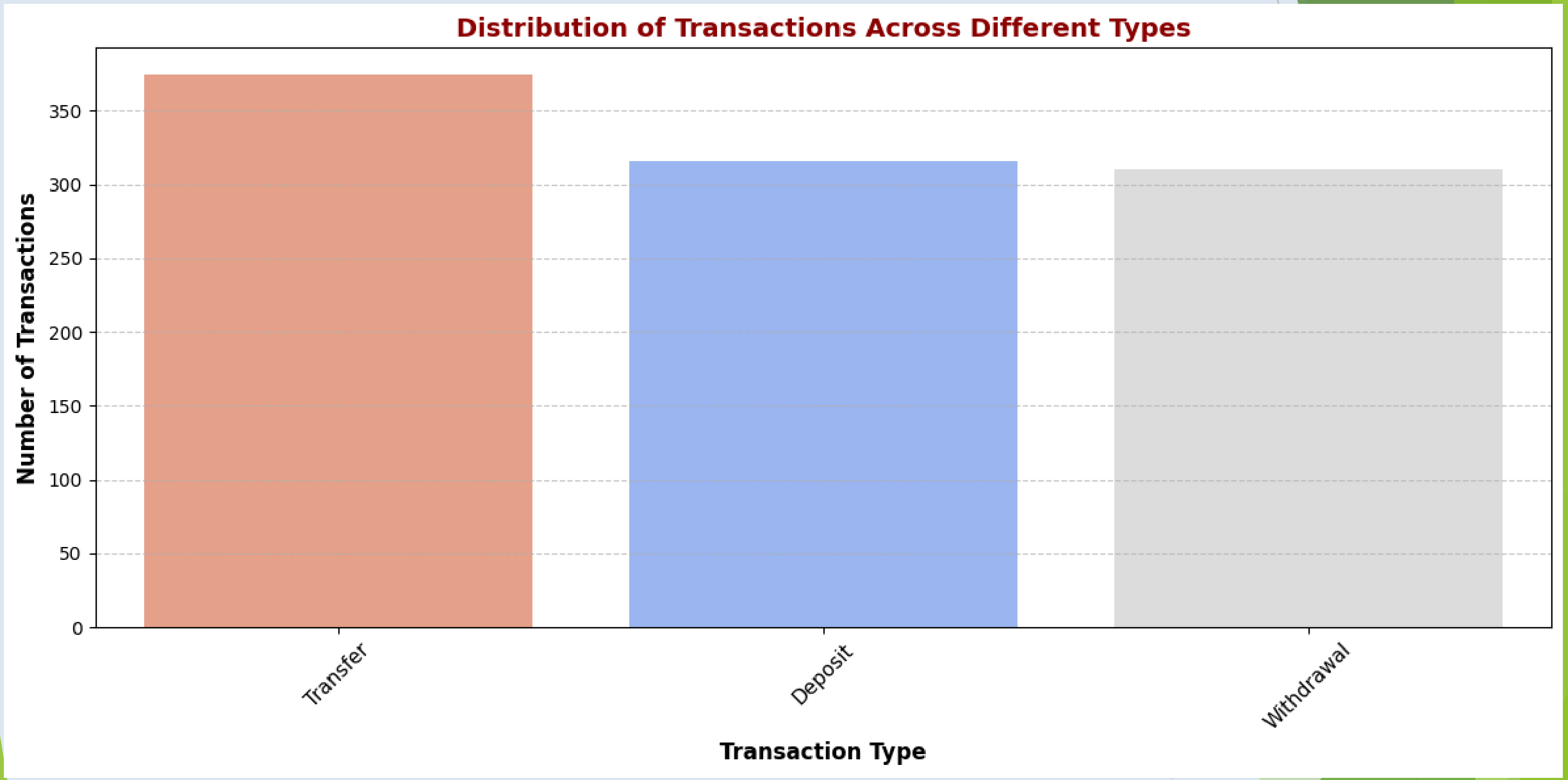
Evaluation & Future Improvements

- - Model confidence depends on dataset quality and completeness
- - Future work: Implement machine learning for real-time fraud detection
- - Improve data collection on network conditions and security measures
- - Conduct further analysis on geographical fraud trends

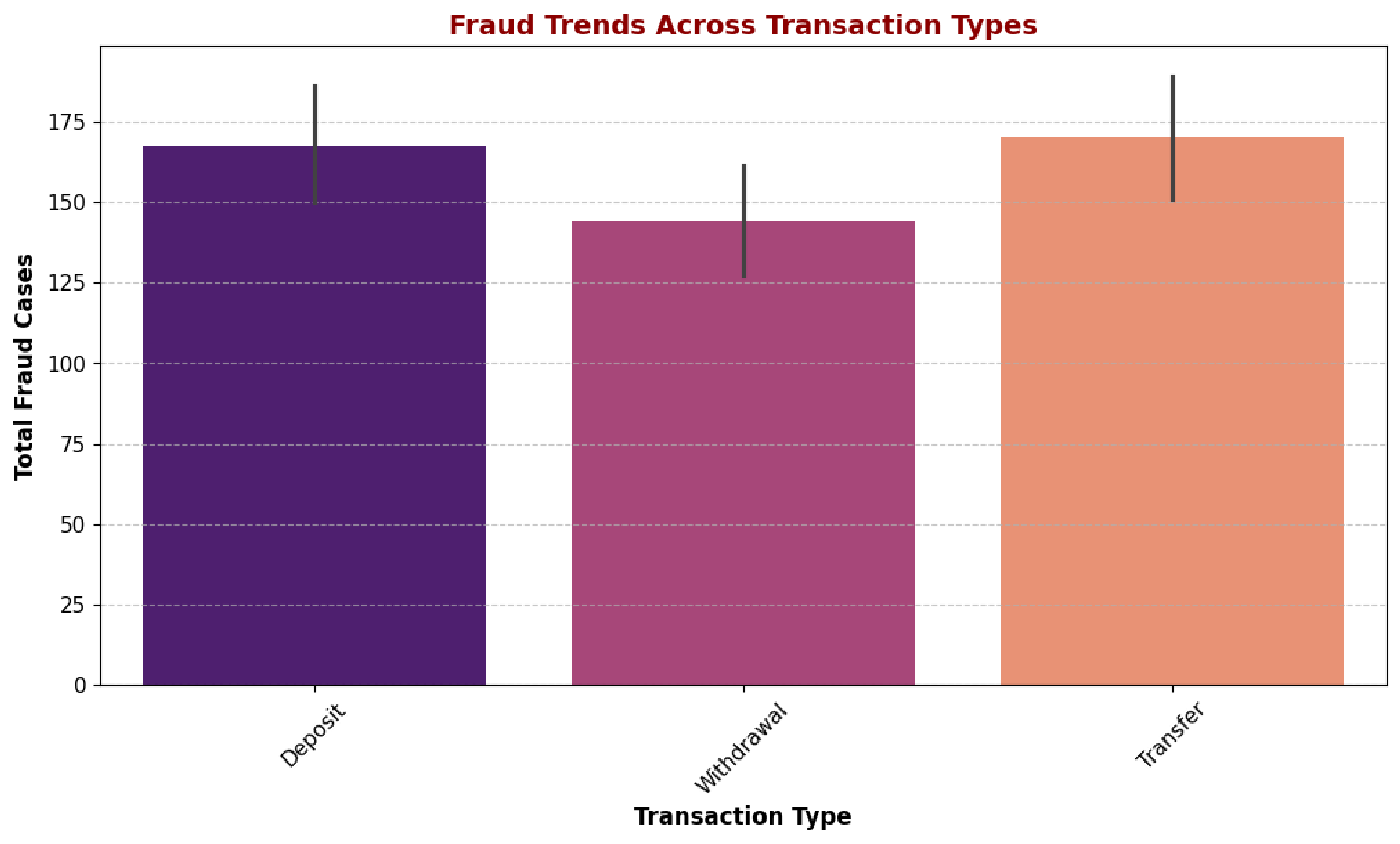
Next Steps

- Implement machine learning models for real-time fraud detection
- Enhance data collection on user behavior and transaction patterns
- Improve network performance analysis to reduce transaction failures
- Collaborate with financial institutions to refine fraud prevention strategies
- Conduct A/B testing on fraud detection methods to evaluate effectiveness

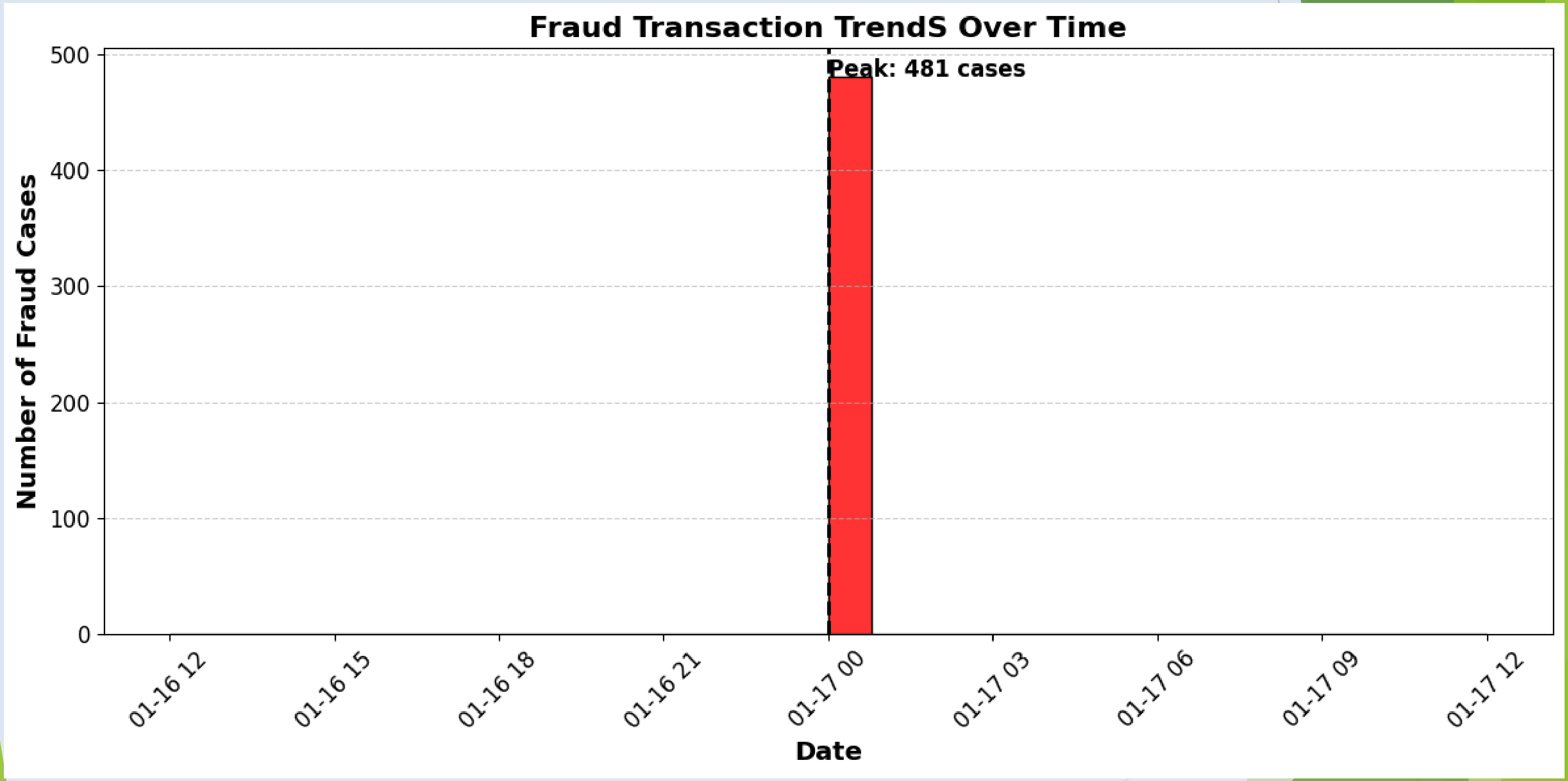
Visualization: Fraudulent Transactions



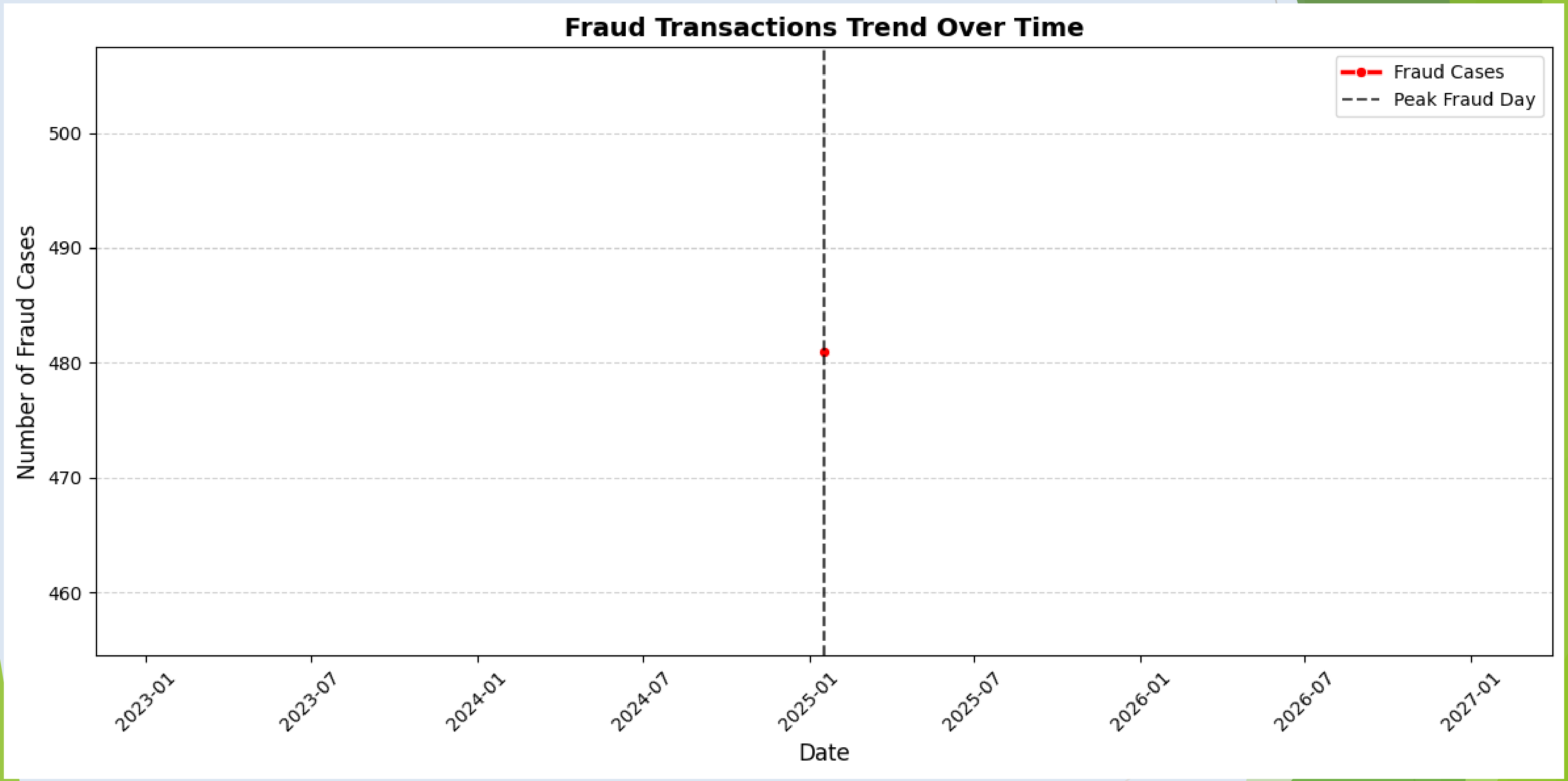
Transaction Success Rate Analysis



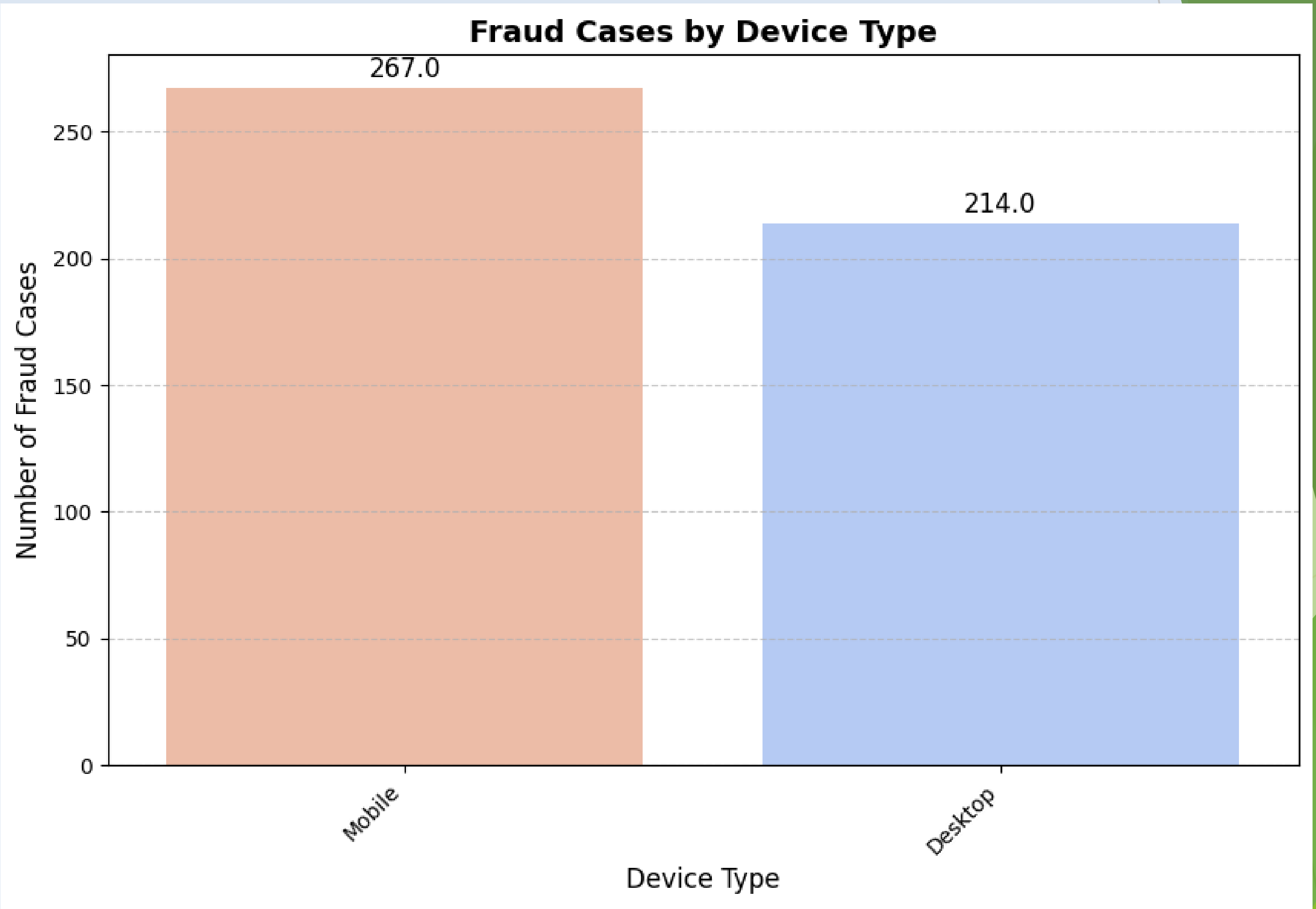
Geographical Distribution of Fraud



Impact of Device Type on Fraud

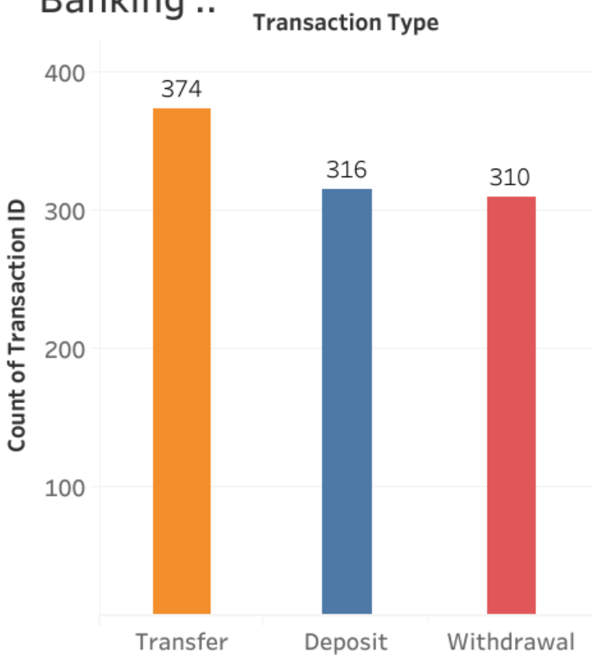


Network Conditions and Transaction Failures

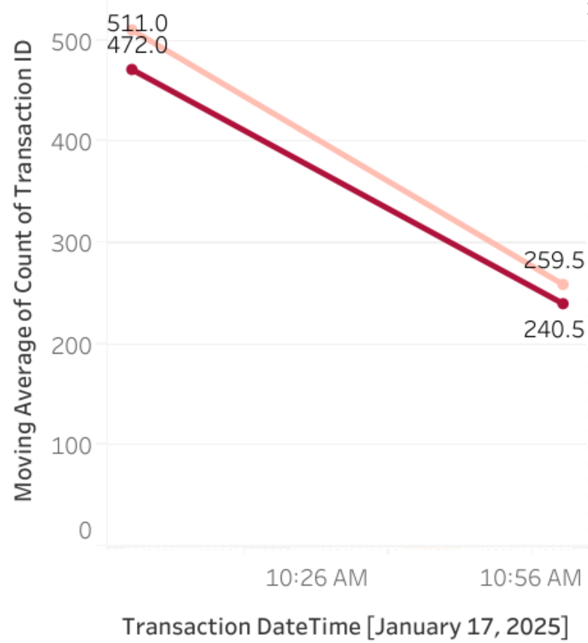


Banking Transaction Analysis: Trends, Fraud & Performance Dashboard

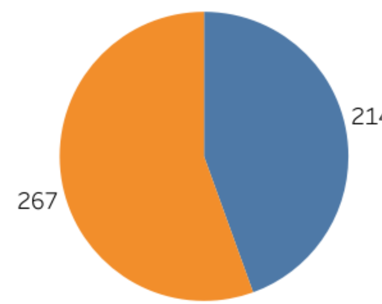
Transaction Type Distribution Banking ..

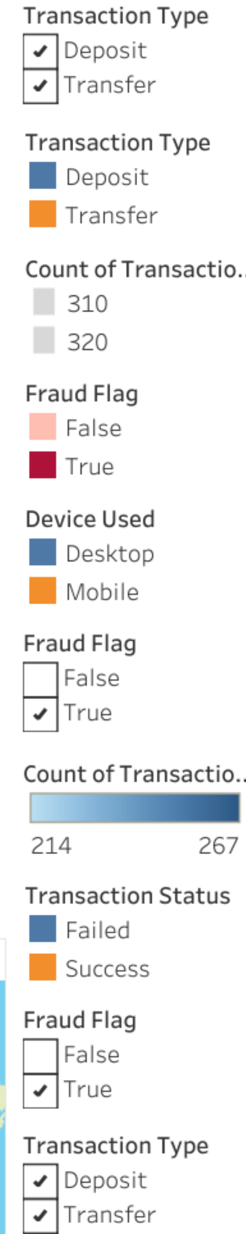


Fraud Trends Over Time

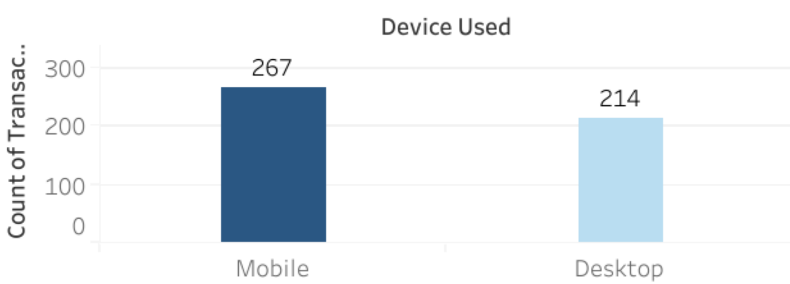


Pie Chart of Fraud Cases by Device Type





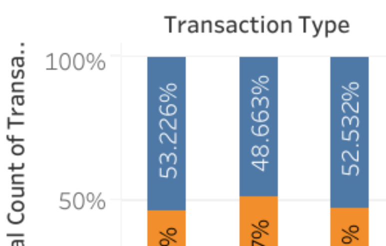
Bar Chart of Fraud Cases by Device Type.



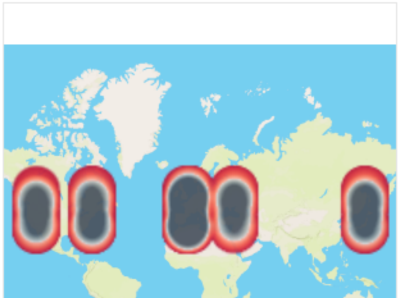
Impact of Latency on Transaction Success

Transaction Status

Stacked Bar Chart for Transaction Success vs. Failure Rate



Geographical Heatmap of Fraud Cases



Insights, Actions, and Future Considerations

- Summary of Key Insights: Fraud detection, transaction efficiency, and data-driven strategies.
- Call to Action: Implement recommended improvements and leverage data insights for better security.
- Future Considerations: Continue refining fraud detection models and optimizing transaction processes.
- Thank You! Questions & Discussions Welcome.

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